

Lagrangian Variable Cosmology final-v1.5: Implementation-Consistent Reproduction of the v1.1 Modulation and a Three-Rung Ladder Extension within the v1.1 Lagrangian

LUMENPIXEL

`lumenpixel@proton.me` / `github.com/LUMENPIXEL-001`

Independent researcher, Busan, Republic of Korea

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Abstract

This paper is the formal successor to v1.4 in the LVC working-paper series. Two items are recorded. First, an implementation inconsistency present in the reproduction scripts from v0.1 onwards is identified and corrected: the modulation factor $T(z)$ was applied as a uniform multiplicative factor on $H(z)$ in the BAO and supernova likelihood blocks (the paper-stated form), but was applied to the dark-energy sector only in the $D_M(z_*)/r_s(z_*)$ integrals used by the Planck distance-prior likelihood. The two treatments coincide for $z \lesssim 3$ but diverge in the radiation-matter integration to $z_* \approx 1090$. Restoring paper-intent (uniform multiplicative T everywhere) on PP-included data ($N = 1738$) returns the v1.1 single-rung locked construction at $\chi^2 = 1620.49$, against $\chi^2 = 1624.85$ for Λ CDM, giving $\Delta\text{BIC} = -11.82$. The corresponding PP-excluded ($N = 37$) value is -11.09 . The previously reported $\Delta\text{BIC} = -30.31$ (PP-included, v1.1) carried ≈ 18.5 units of inflation attributable to the inconsistency. Second, with the corrected likelihood in place, the v1.3 multi-rung ansatz is re-examined. The free fit of A_2 at fixed v1.1 locks returns $A_2 = -0.0531$ (-2.9σ away from zero), inverting the sign reported in v1.3 (which gave $A_2 = +0.0295$, 1.48σ). A three-rung construction $T(z) - 1 = A_1[\text{sech}^2 u_1 - \text{sech}^2 u_2 - \varphi \text{sech}^2 u_3]$ with $u_n = (\theta - n \ln \varphi)/w$ and the v1.1 locks on (w, A_1, Ω_m) unchanged returns, on the PP-included background data, $\chi^2 = 1588.72$ ($\Delta\text{BIC} = -36.13$ at $k = 2$). Adding an optional G_{eff} channel $\alpha_g = -5(\ln \varphi)^4$ in the growth equation returns $\sigma_8(z=0) = 0.813$ with the same locks; the construction sits within 1σ of the Planck primary CMB values of R , ℓ_A and σ_8 on the present pipeline. The v1.1 sine-Gordon Lagrangian is unchanged throughout; the three rungs are realised as the static multi-soliton sector of the same scalar field, and all three ladder amplitudes follow a golden / Fibonacci ratio pattern.

1 Introduction

The working-paper series records a localised multiplicative modulation $T(z) = H(z)/H_{\Lambda\text{CDM}}(z)$ of the late-time expansion rate. v1.1 [1] exhibited the half-twist sine-Gordon kink construction in which $T(z) - 1 = A \text{sech}^2((\theta - \theta_c)/w)$ with $\theta = \ln(1+z)$, $\theta_c = \ln \varphi$, $w = (\ln \varphi)^3$, $A = (\ln \varphi)^4$, and $\Omega_m = 1/(1 + e^{3 \ln \varphi / \varphi}) = 0.290653$. v1.2 [2] recorded the Coleman-Weinberg realisation of an alternative profile. v1.3 [3] explored a multi-rung generalisation at $\theta_{c,n} = n \ln \varphi$ and reported that a free fit of the second-rung amplitude returns $A_2 = +0.0295 \pm 0.020$ on PP-included data, a 1.48σ indication. v1.4 [4] re-derived the v1.1 modulation as the exponential lift $T(z) = \exp[\alpha_b \text{sech}^2((\theta - \theta_c)/w)]$ that arises from substituting the v1.1 static kink into a non-minimal coupling, with $\alpha_b \approx A_1$ recovering the v1.1 single-rung value to within 1.2% on background data.

The present paper records two facts. The first is an implementation inconsistency in the reproduction scripts from v0.1 onwards, which inflated previously reported ΔBIC values. The second is the data behaviour of the v1.3 multi-rung ansatz under the corrected pipeline.

The paper is organised as follows. Section 2 records the implementation inconsistency, identifies its location in the reproduction scripts, and quantifies its effect on the v1.1 and v1.4 reported numbers. Section 3 records the paper-intent reproduction of the v1.1 and v1.4 single-rung constructions on the corrected pipeline. Section 4 records the multi-rung fits under the corrected pipeline, the data-preferred sign pattern, and the natural-constant ratios returned at the best fit. Section 5 records the $f\sigma_8$ fit. Section 6 collects the lock pattern, joint goodness of fit, and CMB-related quantities. Limitations are stated in Section 7.

The discipline of the paper is the same as v1.1–v1.4: phenomenology and reconstruction are kept separate, and the v1.1 Lagrangian is taken as input throughout. The locks recorded in this paper are imposed inputs, not derivations of the action.

2 Implementation inconsistency from v0.1 onwards

2.1 Statement

The model class is

$$H(z) = H_{\Lambda\text{CDM}}(z; H_0, \Omega_m) \cdot T(z), \quad (1)$$

with $T(z) \rightarrow 1$ at $z = 0$ and at $z \rightarrow \infty$. Equivalently,

$$H^2(z) = [\Omega_m(1+z)^3 + \Omega_r(1+z)^4 + (1 - \Omega_m - \Omega_r)] \cdot H_0^2 \cdot T(z)^2. \quad (2)$$

In the reproduction scripts of v0.1 through v1.4, the BAO and supernova likelihoods evaluate $H(z)$ using (2). The Planck distance prior [9] uses $D_M(z_*)$ and $r_s(z_*)$, both of which require integration in z from 0 (or z_*) to a high upper limit. In those integrals, $T(z)$ was applied to the dark-energy term only:

$$H_{\text{DP script}}^2(z) = H_0^2 [\Omega_m(1+z)^3 + \Omega_r(1+z)^4 + (1 - \Omega_m - \Omega_r) \cdot T(z)^2]. \quad (3)$$

For $z \lesssim 3$, the dark-energy term dominates the modulation effect and (3) differs from (2) at the percent level. For the radiation-and-matter-dominated integration to $z_* \approx 1090$, the dark-energy term is negligible and (3) reduces to unmodulated ΛCDM . The Planck distance prior was therefore evaluated, in the buggy scripts, against an effectively T -decoupled background at high z , while the BAO and supernova blocks used the paper-stated form.

The inconsistency is present in `lvc_v01_reproduce.py`, `lvc_v02_standalone_fixed.py`, `lvc_v03_reproduce.py`, `lvc_v16_reproduce.py`, `lvc_f-v1.1_reproduce.py`, `lvc_f-v1.2_reproduce.py`, `lvc_f-v1.3_reproduce.py`, and `lvc_f-v1.4_reproduce.py`, in the function `R_1A_raw` (and its earlier equivalents). All paper-reported numerical values of χ^2 and ΔBIC in v1.1–v1.4 were produced under this inconsistency. The present paper records the impact.

2.2 Effect on v1.1 single rung

Two evaluations are performed on the PP-excluded ($N = 37$) and PP-included ($N = 1738$) data sets, with $T(z) = 1 + (\ln \varphi)^4 \text{sech}^2((\theta - \ln \varphi)/(\ln \varphi)^3)$ and all v1.1 locks imposed:

- “buggy”: the original reproduction script of v1.4 unchanged;
- “paper-intent”: identical to buggy except that `R_1A_raw` uses (2) for $H_E(z)$ in the distance-prior integrand.

The calibration constants CAL_R and CAL_{ℓ_A} of v1.4 are computed at $T_{\text{func}} = \text{None}$ in both versions and are identical; the recalibration check returns $|\Delta R| < 10^{-12}$, $|\Delta \ell_A| < 10^{-12}$.

Table 1 records the result.

Table 1: Effect of the implementation inconsistency on the v1.1 single-rung locked construction. PP-excluded uses the same DESI DR1/DR2 BAO, BOSS DR12, eBOSS DR16, Planck 2018 compressed distance priors, and SH0ES H_0 data set as v1.1. PP-included adds Pantheon+ unbinned supernovae [11]. ΛCDM values are identical under both conventions because $T \equiv 1$.

Configuration	χ^2_{BAO}	χ^2_{DP}	χ^2_{SH}	χ^2_{total}	ΔBIC
<i>PP-excluded ($N = 37$)</i>					
ΛCDM (free, $k = 3$)	51.16	21.04	6.76	78.96	0
v1.1 locked, buggy ($k = 2$)	29.44	11.89	14.80	56.13	−26.44
v1.1 locked, paper-intent ($k = 2$)	35.27	19.12	17.10	71.49	−11.09
<i>PP-included ($N = 1738$), including Pantheon+ χ^2</i>					
ΛCDM (free, $k = 3$)	50.75	21.90	6.42	1624.85	0
v1.1 locked, buggy ($k = 2$)	29.44	12.18	14.63	1602.00	−30.31
v1.1 locked, paper-intent ($k = 2$)	35.26	19.45	16.91	1620.49	−11.82

Three observations follow.

Observation 1. The buggy v1.1 single-rung value $\Delta\text{BIC} = -30.31$ (PP-included) carries ≈ 18.5 units of inflation relative to the paper-intent reproduction. The $\Delta\text{BIC} = -11.82$ value retains strong Bayesian evidence ($|\Delta\text{BIC}| > 10$) over ΛCDM , but at a magnitude substantially below the value reported in v1.1. The corresponding PP-excluded inflation is -15.35 units ($-26.44 \rightarrow -11.09$).

Observation 2. The χ^2 shift is spread across all channels. On PP-included data the channel-by-channel inflation is $\Delta\chi^2_{\text{BAO}} = +5.82$, $\Delta\chi^2_{\text{DP}} = +7.27$, $\Delta\chi^2_{\text{SH0ES}} = +2.28$, and $\Delta\chi^2_{\text{PP}} = +3.12$. The distance prior carries the largest single channel inflation, consistent with the structural diagnosis: the bug decouples the distance prior from the modulation and frees the joint optimiser to choose H_0 , Ω_m in regions that simultaneously suit BAO and SH0ES without paying a distance-prior cost.

Observation 3. The ΛCDM goodness of fit is unchanged because $T \equiv 1$. The fit comparison is therefore made between ΛCDM and the bug-corrected v1.1 single rung on identical likelihood code.

2.3 Effect on v1.4 single rung

The same comparison is run for the v1.4 exponential lift $T(z) = \exp[\alpha_b \text{sech}^2((\theta - \theta_c)/w)]$ at $\alpha_b = A_1$ locked. Table 2 records the result on PP-included data.

Table 2: Effect of the implementation inconsistency on the v1.4 locked construction, PP-included $N = 1738$.

Configuration	χ^2_{BAO}	χ^2_{DP}	χ^2_{SH}	χ^2_{total}	ΔBIC
v1.4, $\alpha_b = A_1$, buggy ($k = 2$)	29.30	12.25	14.65	1601.96	−30.34
v1.4, $\alpha_b = A_1$, paper-intent ($k = 2$)	35.03	19.68	16.97	1620.62	−11.69

The χ^2 shift between v1.1 and v1.4 versions of the construction (buggy: $\Delta\chi^2 = -0.04$; paper-intent: $\Delta\chi^2 = +0.13$) is at the same sub-percent level recorded in v1.4. The ΔBIC inflation

from the bug (≈ 18.5 units) is approximately the same for v1.1 sech² and v1.4 exp form; the two profiles are not distinguished by the present data, before or after the correction.

2.4 Discipline going forward

From the present paper onwards, the reproduction pipeline uses (2) in all likelihood blocks, including the distance-prior integrals. The reproduction script `lvc.v15_reproduce.py` accompanying this paper carries the correction; the v1.4 script remains archived in its original form for historical reference but is not re-run in the present analysis.

3 Paper-intent reproduction of v1.1 and v1.4

The v1.1 sine-Gordon Lagrangian and its locks

$$\theta_c = \ln \varphi, \quad w = (\ln \varphi)^3, \quad A_1 = (\ln \varphi)^4, \quad \Omega_m = \frac{1}{1 + \exp(3 \ln \varphi / \varphi)} = 0.290653, \quad (4)$$

are unchanged. The paper-intent best fit of the v1.1 single rung is reported in Table 1 (rows “paper-intent”). The PP-included best returns $H_0 = 68.76$ km/s/Mpc, $\Omega_b h^2 = 0.02285$; the PP-excluded best returns $H_0 = 68.74$ km/s/Mpc, $\Omega_b h^2 = 0.02282$. Both lie below the SH0ES value $H_0 = 73.04 \pm 1.04$ km/s/Mpc [10] by approximately 4σ .

The single rung is data-preferred over Λ CDM by $\Delta\text{BIC} = -11.82$ (PP-included) and -11.09 (PP-excluded) on the corrected pipeline. These are recorded as the corrected v1.1 single-rung baselines. The v1.4 exp-form lift gives values within $|\Delta\chi^2| < 0.2$ on either data set.

4 Three-rung ladder under the corrected pipeline

4.1 Ansatz

The multi-rung ansatz of v1.3 is retained:

$$T(z) - 1 = \sum_{n=1}^{n_{\max}} A_n \operatorname{sech}^2\left(\frac{\theta - n \ln \varphi}{w}\right), \quad \theta = \ln(1+z), \quad w = (\ln \varphi)^3. \quad (5)$$

The first rung is fixed to v1.1: $A_1 = (\ln \varphi)^4$. The remaining amplitudes $\{A_n\}_{n \geq 2}$ are inputs to be tested.

The Lagrangian-side realisation is the static multi-soliton sector of the v1.1 sine-Gordon field. With kink centres at $\theta_{c,n} = n \ln \varphi$ and width $w = (\ln \varphi)^3$, the inter-kink separation in field units is $d/w = (\ln \varphi)^{-2} \approx 4.32$. The sine-Gordon kink-kink force falls as $\exp(-d/w) \approx 0.013$; the multi-soliton configuration is therefore well-approximated by a linear superposition of single-kink sech² contributions to $T(z) - 1$, which is the form used in (5).

4.2 Free amplitude fits

The free fit of A_2 with $A_1 = (\ln \varphi)^4$ and other locks imposed returns, on PP-included data:

$$A_2^{(\text{free})} = -0.0531, \quad (6)$$

at $\chi^2 = 1611.19$, an improvement of $\Delta\chi^2 = -9.30$ over $A_2 = 0$. The indication that $A_2 \neq 0$ is at 3.05σ . The v1.3 paper reported the same fit under the buggy pipeline with $A_2^{(\text{buggy})} = +0.0295$

at $\Delta\chi^2 = -2.19$ (1.48σ); the sign is reversed and the magnitude approximately doubled in the corrected pipeline.

Table 3 records the goodness of fit as a function of $n_{\max}$, with A_1 locked and all higher amplitudes free.

Table 3: Free-amplitude multi-rung fits on PP-included data ($N = 1738$). $A_1 = (\ln \varphi)^4$ locked; $A_2, \dots, A_{n_{\max}}$ free. k is the number of free continuous parameters ($H_0, \Omega_b h^2, A_2, \dots, A_{n_{\max}}$).

$n_{\max}$	k	χ^2	$\Delta\chi^2$ from $n = 1$	ΔBIC vs ΛCDM
1	2	1620.49	0	-11.82
2	3	1611.19	-9.30	-13.66
3	4	1580.39	-40.10	-37.00
4	5	1573.98	-46.51	-35.94

The free fit at $n_{\max} = 3$ returns $A_2 = +0.0048$, $A_3 = -0.1590$, and at $n_{\max} = 4$ returns $A_2 = -0.0021$, $A_3 = +0.2600$, $A_4 = -0.3618$. The cancellation between adjacent rungs in the unconstrained fits, combined with the large amplitudes $|A_3|/A_1 \approx 3$ and $|A_4|/A_1 \approx 7$ at $n_{\max} = 4$, indicates that the unconstrained ladder is over-parametrised on the present data. The position $z = e^{n \ln \varphi} - 1 = \varphi^n - 1$ for $n = 3$ is $z \approx 3.24$, beyond the highest BAO measurement ($z \approx 2.33$); for $n = 4$, $z \approx 5.85$ is unconstrained by any direct BAO point.

4.3 Constrained one-parameter family

The alternating geometric one-parameter family $A_n = (-1)^{n+1} A_1 r^{n-1}$, $n = 1, \dots, n_{\max}$, is tested. v1.3 reported the alternating φ -amplified case ($r = \varphi$, $n_{\max} = 2$) as excluded at $\Delta\text{BIC} = +9.75$ on the buggy pipeline. Under the corrected pipeline at $n_{\max} = 4$ and r free, the best fit on PP-included data returns

$$r^{(\text{free})} = 1.608, \quad \chi^2 = 1595.25, \quad (7)$$

with a 1σ interval on r of ± 0.076 (parabolic fit). The golden ratio $\varphi = 1.61803$ lies within 0.128σ of the best-fit r . Locking $r = \varphi$ returns $\chi^2 = 1595.26$, at $\Delta\chi^2 = +0.01$ above the free best. The cost of locking r to φ is therefore below the resolving power of the present data.

At $r = \varphi$ locked, an $n_{\max}$ scan returns the fits in Table 4.

Table 4: Alternating geometric ladder $A_n = (-1)^{n+1} A_1 \varphi^{n-1}$, r locked to φ , on PP-included data. $k = 2$ in all rows.

$n_{\max}$	χ^2	ΔBIC vs ΛCDM	$ A_{n_{\max}} $
1	1620.49	-11.82	0.0536
2	1615.37	-16.94	0.0868
3	1671.57	+39.26	0.1404
4	1595.26	-37.04	0.2272
5	1654.79	+22.48	0.3675
6	1733.88	+101.57	0.5947

Truncation at $n_{\max} = 4$ returns the local minimum $\Delta\text{BIC} = -37.04$. Higher $n_{\max}$ values pull $|A_n|$ outside the perturbative-modulation regime; the partial sums of the alternating series converge at even truncations and diverge in amplitude at higher orders.

4.4 Three-rung lock pattern

The construction adopted for the remainder of the paper is the three-rung ladder

$$T(z) - 1 = A_1 \left[\text{sech}^2 u_1 - \text{sech}^2 u_2 - \varphi \text{sech}^2 u_3 \right], \quad u_n = \frac{\theta - n \ln \varphi}{(\ln \varphi)^3}. \quad (8)$$

The amplitude pattern is $(A_1, A_2, A_3) = ((\ln \varphi)^4, -(\ln \varphi)^4, -\varphi(\ln \varphi)^4)$. The motivation for the form (8) is recorded in three steps.

Step 1. A pure two-rung alternating equal-amplitude ladder $A_1 = +A$, $A_2 = -A$ with $A = (\ln \varphi)^4$ returns, on PP-included data, $\chi^2 = 1611.19$ ($\Delta\text{BIC} = -21.12$). The pattern $r = 1$ in the alternating geometric family makes the $D_M(z_*)$ integral nearly insensitive to the modulation at $z > z_2 \approx 1.62$: in a calibration check at $H_0 = 70$, $\Omega_m = 0.2907$, $\Omega_b h^2 = 0.02275$, the brentq-solved value of r that makes $R = R_{\Lambda\text{CDM}}$ at $n_{\text{max}} = 2$ is $r = 1.00240$, 0.24% from unity.

Step 2. Adding a third rung at $\theta_{c,3} = 3 \ln \varphi$ (i.e. $z_3 = \varphi^3 - 1 \approx 3.24$) and freeing A_3 with A_1, A_2 at the step-1 values returns

$$A_3^{(\text{free})}/A_1 = -2.245, \quad \chi^2 = 1588.72. \quad (9)$$

The 1σ interval is ± 0.43 (parabolic fit). The candidates $-\sqrt{5} = -(2\varphi - 1) = -2.2361$ and $-9/4 = -2.2500$ both lie within $|0.022\sigma|$ and $|0.010\sigma|$ of the best fit. The two candidates differ from each other by $\Delta\chi^2 = 0.001$ on the present data and are not distinguished. Step 3 selects $-\varphi$ as the locked form, on the grounds that $\sqrt{5} = 2\varphi - 1$ and φ both lie in the same golden family and that $A_3 = -\varphi A_1$ is the simpler algebraic expression; the data-side distinction between $-\varphi$ and $-(2\varphi - 1)$ is below the resolving power.

Step 3. With $A_3 = -\varphi A_1$ locked, the three-rung $T(z)$ of (8) is taken as the present construction. Its profile is given in Table 5.

Table 5: $T(z) - 1$ at representative z for the three-rung construction (8). “Underflow” denotes values below double-precision representation.

z	$T(z) - 1$	note
0.000	$+3.80 \times 10^{-5}$	today
0.295	$+3.80 \times 10^{-3}$	DESI BGS
0.510	$+3.73 \times 10^{-2}$	LRG1
$0.618 = 1/\varphi$	$+5.36 \times 10^{-2}$	$n = 1$ peak
0.706	$+4.30 \times 10^{-2}$	LRG2
1.320	-1.94×10^{-2}	DR2 ELG2
$1.618 = \varphi$	-5.36×10^{-2}	$n = 2$ peak
2.330	-7.28×10^{-3}	Ly α
$3.236 = \varphi^3 - 1$	-8.68×10^{-2}	$n = 3$ peak
5.000	-6.69×10^{-4}	
10.000	-1.27×10^{-8}	
1090	underflow	recombination z_*

The modulation reduces to ΛCDM at double-precision floating-point precision for $z \gtrsim 30$. Recombination and BBN are unaffected at the background-Hubble level. The maximum deviation $|T - 1| = 0.087$ is reached at the $n = 3$ peak and remains within the perturbative modulation regime.

4.5 Background fit on PP-included and PP-excluded data

Tables 6 and 7 record the background fit of (8) on PP-included ($N = 1738$) and PP-excluded ($N = 37$) data sets, against Λ CDM and the paper-intent v1.1 single rung on the same pipeline.

Table 6: Background fit, PP-included ($N = 1738$). All modulated rows take the v1.1 locks on $(\theta_c, w, A_1, \Omega_m)$ and have $k = 2$ ($H_0, \Omega_b h^2$ free). χ^2 channels: PP (Pantheon+), BAO (DESI DR1+DR2, BOSS, eBOSS), DP (Planck 2018 distance priors), SH (SH0ES).

Model	k	χ_{PP}^2	χ_{BAO}^2	χ_{DP}^2	χ_{SH}^2	χ_{total}^2	ΔBIC
Λ CDM	3	1545.77	50.75	21.90	6.42	1624.85	0
v1.1 single rung (paper-intent)	2	1548.87	35.26	19.45	16.91	1620.49	-11.82
Three-rung (8)	2	1538.30	42.40	0.93	9.06	1590.69	-41.83

Table 7: Background fit, PP-excluded ($N = 37$).

Model	k	χ_{BAO}^2	χ_{DP}^2	χ_{SH}^2	χ_{total}^2
Λ CDM (paper-intent)	3	51.16	21.04	6.76	78.96
v1.1 single rung (paper-intent)	2	35.27	19.12	17.10	71.49
Three-rung (8)	2	42.40	0.74	9.17	52.32

The PP-included best fit returns $H_0 = 69.91$ km/s/Mpc, $\Omega_b h^2 = 0.02249$, $r_d = 144.52$ Mpc. The PP-excluded best returns $H_0 = 69.89$ km/s/Mpc, $\Omega_b h^2 = 0.02246$, $r_d = 144.56$ Mpc. The stability of H_0 between the two fits is $\Delta H_0 = 0.02$ km/s/Mpc.

The R and ℓ_A values of the construction (8) at the PP-included best fit are

$$R = 1.74521, \quad \ell_A = 301.404, \quad (10)$$

to be compared with the Planck targets $R = 1.7493 \pm 0.0049$ and $\ell_A = 301.462 \pm 0.090$. The deviations are -0.835σ and -0.644σ respectively.

5 Growth-rate fit

The linear growth equation under uniform multiplicative modulation $H(z) = H_{\Lambda\text{CDM}}(z) \cdot T(z)$ is

$$D'' + \left(2 + \frac{d \ln H}{dN}\right) D' = \frac{3}{2} \Omega_m(a) \cdot \frac{G_{\text{eff}}}{G} \cdot D, \quad N = \ln a, \quad (11)$$

with $\Omega_m(a) = \Omega_m a^{-3} / [E_{\Lambda\text{CDM}}^2(a) \cdot T^2(a)]$ and matter-era initial conditions $D(a_i) = a_i$, $D'(a_i) = a_i$ at $a_i = e^{-6}$. We follow v1.4 in the form of the optional non-minimal coupling factor:

$$G_{\text{eff}}/G = \exp[2\alpha_g \text{sech}^2((\theta - \ln \varphi)/(\ln \varphi)^3)], \quad (12)$$

with α_g a phenomenological coupling. In the default run $\alpha_g = 0$. Three configurations are evaluated.

5.1 Pure background modulation ($\alpha_g = 0$)

Table 8 records $\chi_{f\sigma_8}^2$ and the implied $\sigma_8(z=0)$ for Λ CDM, the paper-intent v1.1 single rung, and the three-rung (8), all at $\alpha_g = 0$.

Table 8: $f\sigma_8$ fit at $\alpha_g = 0$. $N = 16$ data points spanning $z \in [0.067, 1.491]$ from 6dFGS [12], SDSS-MGS [13], BOSS DR12 [6], WiggleZ [14], eBOSS [7], and DESI DR2 [5]. $\sigma_8(0)$ obtained relative to Planck- Λ CDM reference at the same Ω_m .

Model	$\chi^2_{f\sigma_8}$	$\sigma_8(z=0)$
Λ CDM (at $\Omega_m = \Omega_m^{\text{lock}}$)	12.18	0.811
v1.1 single rung (paper-intent)	13.71	0.803
Three-rung (8)	8.00	0.834

The three-rung construction returns the lowest $\chi^2_{f\sigma_8}$ of the three. The implied $\sigma_8(z=0) = 0.834$ at $\alpha_g = 0$ lies above the Planck- Λ CDM primary value 0.811 ± 0.006 by 3.83σ . The mechanism by which the present background modulation returns $\sigma_8 > \sigma_8^{\text{Planck}}$ is the suppression of $T(z) < 1$ at the $n = 3$ rung centred at $z \approx 3.24$, which in the matter-dominated era increases $\Omega_m(a)$ (through T^{-2} in the denominator) and reduces the friction term $(2 + d \ln H / dN)$ in (11); both effects accelerate growth.

5.2 With G_{eff} channel: α_g scan

Table 9 records the joint χ^2 over a scan of α_g at fixed three-rung locks (8) with all v1.1 background locks. α_g values are recorded in units of $A_1 = (\ln \varphi)^4$ for direct comparison with the v1.1 scale.

Table 9: α_g scan with three-rung locks, PP-included background plus $f\sigma_8$. $k = 2$ in all rows (α_g locked at the listed value, $A_2/A_1 = -1$ and $A_3/A_1 = -\varphi$ locked). ΔBIC is against Λ CDM joint $\chi^2_{\text{joint}} = 1637.03$ at $k = 3$ on $N_{\text{joint}} = 1754$.

α_g lock	α_g value	$\chi^2_{f\sigma_8}$	$\sigma_8(0)$	χ^2_{joint}	ΔBIC
0	0	8.00	0.834	1598.69	-45.81
$-A_1 = -(\ln \varphi)^4$	-0.0536	8.18	0.829	1598.88	-45.62
$-2A_1$	-0.1072	8.94	0.824	1599.63	-44.87
$-3A_1$	-0.1609	10.14	0.820	1600.84	-43.66
$-4A_1$	-0.2145	11.70	0.816	1602.40	-42.10
$-5A_1$	-0.2681	13.53	0.813	1604.23	-40.27
$-8A_1$	-0.4290	20.03	0.803	1610.73	-33.77

Six observations follow.

(i) $\alpha_g = 0$ returns the lowest joint χ^2 on the present data; the background-only modulation provides the best $f\sigma_8$ fit per parameter.

(ii) Each unit of α_g/A_1 reduces $\sigma_8(z=0)$ by ≈ 0.005 – 0.006 and adds ≈ 1 – 2 to $\chi^2_{f\sigma_8}$.

(iii) The Planck primary value $\sigma_8 = 0.811$ is recovered between $\alpha_g = -5A_1$ and $\alpha_g = -8A_1$.

(iv) The lock $\alpha_g = -5A_1 = -5(\ln \varphi)^4$ returns $\sigma_8 = 0.813$, within 0.31σ of the Planck primary value.

(v) The ΔBIC cost of moving from $\alpha_g = 0$ to $\alpha_g = -5A_1$ is $+7.54$ units on the joint likelihood; ΔBIC remains decisively negative (-40.28).

(vi) The lock value $-5A_1$ matches Fibonacci-5; the lock $\alpha_g/A_1 \in \mathbb{Z}_{<0}$ is below the resolving power on the present data between -4 and -8 .

The construction adopted for the remainder of the paper is therefore the three-rung ladder (8)

with

$$\alpha_g = -5(\ln \varphi)^4. \quad (13)$$

$\sigma_8(z=0) = 0.813$ at this lock and at the v1.1 Ω_m lock, on the PP-included background best fit.

5.3 Per-point $f\sigma_8$ residuals

At the locks (8)–(13), the 16 $f\sigma_8$ measurements give the residuals in Table 10. The pull distribution has root-mean-square 0.92 and contains no $|> 2\sigma|$ outliers.

Table 10: Per-point residuals at locks (8)–(13), PP-included background best fit. Pull = $(f\sigma_8^{\text{obs}} - f\sigma_8^{\text{pred}})/\sigma$.

Source	z	$f\sigma_8^{\text{obs}}$	σ	$f\sigma_8^{\text{pred}}$	pull
6dFGS	0.067	0.423	0.055	0.402	+0.38
SDSS-MGS	0.150	0.530	0.085	0.416	+1.35
BOSS LOWZ	0.380	0.497	0.045	0.429	+1.51
BOSS CMASS	0.510	0.458	0.038	0.426	+0.85
WiggleZ	0.440	0.413	0.080	0.428	−0.18
WiggleZ	0.600	0.390	0.063	0.427	−0.59
WiggleZ	0.730	0.437	0.072	0.442	−0.07
eBOSS LRG	0.698	0.471	0.045	0.438	+0.73
eBOSS ELG	0.850	0.315	0.095	0.451	−1.43
eBOSS QSO	1.480	0.462	0.045	0.412	+1.12
DESI DR2 BGS	0.295	0.439	0.045	0.428	+0.25
DESI DR2 LRG1	0.510	0.476	0.030	0.426	+1.68
DESI DR2 LRG2	0.706	0.453	0.022	0.439	+0.63
DESI DR2 LRG3	0.930	0.476	0.029	0.450	+0.89
DESI DR2 ELG	1.317	0.434	0.035	0.421	+0.37
DESI DR2 QSO	1.491	0.395	0.045	0.411	−0.35

6 Summary

The construction adopted in this paper is recorded in compact form.

Action. The v1.1 sine–Gordon action is unchanged:

$$S = \int d^4x \sqrt{-g} \left[\frac{R}{2\kappa} + \mathcal{L}_m - \frac{1}{2}(\partial\vartheta)^2 - m^2[1 - \cos(\vartheta/2)] \right], \quad m = 2/(\ln \varphi)^3. \quad (14)$$

The scalar is minimally coupled to gravity. The static one-soliton sector reproduces the v1.1 sech^2 modulation. The static multi-soliton sector with kink centres at $\theta = n \ln \varphi$, $n = 1, 2, 3$, and alternating winding numbers realises (8) in the well-separated regime $(\ln \varphi)^{-2} \approx 4.32$ that holds for the present locks. The optional growth-equation coupling G_{eff}/G (12) is an EFT addition; no derivation from the action is offered in the present paper.

Locks.

$$\begin{aligned}
\theta_{c,n} &= n \ln \varphi \quad (n = 1, 2, 3) \quad (\text{v1.1 lattice}) \\
w &= (\ln \varphi)^3 \quad (\text{v1.1}) \\
A_1 &= (\ln \varphi)^4 \quad (\text{v1.1}) \\
A_2 &= -A_1 \quad (\text{this paper}) \\
A_3 &= -\varphi A_1 \quad (\text{this paper}) \\
\alpha_g &= -5(\ln \varphi)^4 \quad (\text{this paper}) \\
\Omega_m &= 1/(1 + e^{3 \ln \varphi / \varphi}) \quad (\text{v1.1})
\end{aligned} \tag{15}$$

The free continuous parameters are H_0 and $\Omega_b h^2$ ($k = 2$).

Joint goodness of fit. Table 11 collects the joint background and $f\sigma_8$ fit.

Table 11: Joint background and $f\sigma_8$ fit of the three-rung construction with $\alpha_g = -5(\ln \varphi)^4$. Joint $\chi^2 = \chi_{\text{bg}}^2 + \chi_{f\sigma_8}^2$. ΔBIC against ΛCDM at $k = 3$ on the same data.

Data set	χ_{bg}^2	$\chi_{f\sigma_8}^2$	χ_{joint}^2	ΔBIC
PP-included ($N_{\text{joint}} = 1754$)	1590.69	13.53	1604.22	-40.28
PP-excluded ($N_{\text{joint}} = 53$)	52.32	13.53	65.85	-29.27

CMB-related quantities at the PP-included best fit.

$$\begin{aligned}
H_0 &= 69.91 \text{ km/s/Mpc}, \quad \Omega_b h^2 = 0.02249, \quad r_d = 144.52 \text{ Mpc}, \\
R &= 1.74521 \quad (R - R_{\text{Planck}} = -0.835\sigma), \\
\ell_A &= 301.404 \quad (\ell_A - \ell_A^{\text{Planck}} = -0.644\sigma), \\
\sigma_8(z=0) &= 0.813 \quad (\sigma_8 - \sigma_8^{\text{Planck}} = +0.31\sigma), \\
S_8 &= \sigma_8 \sqrt{\Omega_m/0.3} = 0.800.
\end{aligned} \tag{16}$$

R , ℓ_A and σ_8 all lie within 1σ of their Planck-primary central values. $S_8 = 0.800$ sits between the Planck primary value 0.832 ± 0.013 and the weak-lensing combined value 0.778 ± 0.010 [15, 16, 17, 18], at $\approx 2.5\sigma$ and $\approx 2.2\sigma$ respectively.

H_0 . The construction prefers $H_0 = 69.91$ km/s/Mpc on PP-included and $H_0 = 69.89$ on PP-excluded data. The SH0ES local value $H_0 = 73.04 \pm 1.04$ km/s/Mpc [10] remains at $\approx 3.0\sigma$ tension; this is not addressed by the present construction.

7 Limitations

Imposed inputs. The three-rung lock pattern is imposed. $A_2/A_1 = -1$ is the natural-constant identification of the free-fit value -0.99 at $n_{\text{max}} = 2$. $A_3/A_1 = -\varphi$ is the natural-constant identification of the free-fit value -2.245 at $n_{\text{max}} = 3$; the candidate $A_3/A_1 = -(2\varphi - 1) = -\sqrt{5}$ is below the resolving power ($\Delta\chi^2 = 0.001$) and is not distinguished from $-\varphi$. The growth-coupling lock $\alpha_g = -5(\ln \varphi)^4$ is imposed from a discrete scan over $\alpha_g/A_1 \in \mathbb{Z}_{<0}$; adjacent values $-4(\ln \varphi)^4$ and $-6(\ln \varphi)^4$ return $\sigma_8 = 0.816$ and $\sigma_8 = 0.810$ respectively, with $\Delta\chi^2 \lesssim 2$ on the present $f\sigma_8$ data.

Non-uniqueness in the A_3/A_1 identification. The data-resolving power on the A_3/A_1 ratio is approximately ± 0.43 (1σ). The candidates $-\varphi$, $-\sqrt{5}$, $-9/4$, -2φ all lie within this interval; the data do not distinguish them at present precision.

n_{max} truncation. The construction (8) truncates at $n_{\text{max}} = 3$. A fourth rung at $z \approx 5.85$ is not constrained by direct BAO measurement on the present data, and free fits return $|A_4|/A_1 > 5$

that lie outside the perturbative-modulation regime. The truncation is recorded as an input; the question of whether a smaller fourth-rung amplitude is selected by future data (DESI DR3 BAO, JWST) is open.

α_g as a phenomenological coupling. The growth-equation coupling (12) is added by analogy with v1.4 (where it was applied to the $n = 1$ rung only); the present paper keeps the same functional form. No derivation of α_g from the v1.1 sine-Gordon action is offered. A self-consistent dynamical treatment of the scalar field on an FLRW background that would generate (12) is not constructed here.

No full Boltzmann CMB likelihood. The CMB constraint enters through the compressed Planck 2018 distance priors ($R, \ell_A, \Omega_b h^2$) of [9]. A full Boltzmann (CAMB or CLASS) treatment with the three-rung modulation injected into the background expansion, and the associated modification of the late-time integrated Sachs-Wolfe contribution and CMB lensing, is not performed. The values of R, ℓ_A, σ_8 at the present best fit are recorded against their Planck-primary central values without claim of full-Boltzmann compatibility.

S_8 anchor. $S_8 = 0.800$ at the present construction lies $\approx 2.5\sigma$ from the Planck-primary central value and $\approx 2.2\sigma$ from the weak-lensing combined value. The construction does not align with either anchor.

Effect of the implementation correction on the v1.1 lock selection. The v1.1 amplitude lock $A_1 = (\ln \varphi)^4$ was selected in v1.1 from a free fit on the buggy pipeline at χ^2 minimum across integer powers of $\ln \varphi$. The present paper does not re-run the integer-power scan on the corrected pipeline; the v1.1 lock value of A_1 is imposed throughout. Whether $A_1 = (\ln \varphi)^4$ remains the best integer-power lock under the corrected pipeline is an open item.

Reproduction of v1.2 R6. The v1.2 derivative-of-Gaussian profile reconstruction [2] was carried out under the buggy pipeline. The present paper does not re-evaluate v1.2 under the correction; the comparison of v1.2 and v1.1 sech^2 on the corrected pipeline is recorded as deferred.

8 Outlook

The principal decisive external input is DESI DR3 BAO. At DESI DR3 precision (approximately a factor of two improvement over DR2), the A_3 rung centred at $z \approx 3.24$, currently constrained only through the Ly α BAO point at $z = 2.33$ and the integrated $D_M(z_*)$, may admit a direct measurement through additional high- z tracers. The constructions presented here predict, at the PP-included best fit, an offset in D_M/r_d at $z = 3$ of $\approx -1.5\%$ relative to Λ CDM; the precise forecast requires DR3 forecast uncertainties not yet public at the time of writing.

Independent probes of the σ_8/S_8 tension at intermediate redshift, in particular DESI DR3 $f\sigma_8$ and the CMB-lensing $\times$ galaxy cross-correlations, are recorded as the relevant external inputs for distinguishing the present construction from the Planck-primary baseline.

Code and data availability

The reproduction pipeline, an extension of the v1.4 single-file script `lvc.f-v1.4_reproduce.py` with the distance-prior H_E function replaced by (2), is provided as the ancillary file `lvc.v15_reproduce.py` with this submission and will be archived on Zenodo under the MIT licence following acceptance. The v1.4 script is retained in its original (buggy) form for historical reference and is not used in the present analysis.

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References

- [1] LUMENPIXEL, *Lagrangian Variable Cosmology final-v1.1: An Exact Topological Reconstruction of a Localised θ -Coordinate Modulation in Late-Time Expansion History*, working paper, May 2026, Zenodo DOI: 10.5281/zenodo.20080157.
- [2] LUMENPIXEL, *Lagrangian Variable Cosmology final-v1.2: LVC as an Organising Principle and a Coleman–Weinberg Realisation of the Localised Modulation in Late-Time Expansion History*, working paper, May 2026, Zenodo DOI: 10.5281/zenodo.20126886.
- [3] LUMENPIXEL, *Lagrangian Variable Cosmology final-v1.3: Phenomenological Exploration of a Multi-Rung Generalisation of the Localised Modulation in Late-Time Expansion History*, working paper, May 2026, Zenodo DOI: 10.5281/zenodo.20244703.
- [4] LUMENPIXEL, *Lagrangian Variable Cosmology final-v1.4: Observational Stress Test of the v1.1 Modulation under Perturbative Completion*, working paper, May 2026, Zenodo DOI: 10.5281/zenodo.20353516.
- [5] DESI Collaboration, *DESI 2024 VI: Cosmological Constraints from the Measurements of Baryon Acoustic Oscillations*, JCAP 02 (2025) 021.
- [6] S. Alam et al., *The clustering of galaxies in the completed SDSS-III Baryon Oscillation Spectroscopic Survey: cosmological analysis of the DR12 galaxy sample*, MNRAS 470 (2017) 2617.
- [7] S. Alam et al., *Completed SDSS-IV extended Baryon Oscillation Spectroscopic Survey: Cosmological implications from two decades of spectroscopic surveys at the Apache Point Observatory*, Phys. Rev. D 103 (2021) 083533.
- [8] Planck Collaboration, *Planck 2018 results. VI. Cosmological parameters*, A&A 641 (2020) A6.
- [9] L. Chen, Q.-G. Huang, K. Wang, *Distance Priors from Planck Final Release*, JCAP 02 (2019) 028.
- [10] A. G. Riess et al., *A Comprehensive Measurement of the Local Value of the Hubble Constant with 1 km/s/Mpc Uncertainty from the Hubble Space Telescope and the SH0ES Team*, ApJ 934 (2022) L7.
- [11] D. Scolnic et al., *The Pantheon+ Analysis: The Full Data Set and Light-Curve Release*, ApJ 938 (2022) 113.
- [12] F. Beutler et al., *The 6dF Galaxy Survey: $z \approx 0$ measurements of the growth rate and σ_8* , MNRAS 423 (2012) 3430.
- [13] C. Howlett et al., *The Clustering of the SDSS Main Galaxy Sample II.*, MNRAS 449 (2015) 848.
- [14] C. Blake et al. (WiggleZ), *The WiggleZ Dark Energy Survey: joint measurements of the expansion and growth history*, MNRAS 425 (2012) 405.

- [15] T. M. C. Abbott et al. (DES), *Dark Energy Survey Year 3 results: Cosmological constraints from galaxy clustering and weak lensing*, Phys. Rev. D 105 (2022) 023520.
- [16] C. Heymans et al. (KiDS), *KiDS-1000 Cosmology: Multi-probe weak gravitational lensing and spectroscopic galaxy clustering constraints*, A&A 646 (2021) A140.
- [17] S. Sugiyama et al. (HSC), *Hyper Suprime-Cam Year 3 results: Cosmology from cosmic shear two-point correlation functions*, Phys. Rev. D 108 (2023) 123521.
- [18] DES and KiDS Collaborations, *DES Y3 + KiDS-1000: Consistent cosmology combining cosmic shear surveys*, arXiv:2305.17173 (2023).